

Campus Trade Project Documentation

Overview

Project Overview:

The Campus Trade platform enables students to buy and sell used items within their campus community. The project aims to provide a simple and secure marketplace for students to engage in peer-to-peer trade.

Tech Stack:

- Frontend: React.js
- Backend: Node.js, Express
- Database: MongoDB
- Hosting: Heroku

Key Features:

- User authentication (Login/Signup)
- Listing items for sale (Item name, price, description)
- Real-time chat between buyers and sellers
- Search and filter listings
- User ratings and reviews

Tech Stack Used

- Frontend: React.js (for building user interfaces)
- Backend: Node.js (for API handling), Express (for routing)
- Database: MongoDB (for storing user data, listings, etc.)

- Hosting: Heroku (for deployment)

Collection Schema:

1. Users Collection:

- user_id (string, unique)
- username (string)
- email (string)
- password (hashed)
- rating (number)

2. Listings Collection:

- listing_id (string, unique)
- user_id (foreign key referencing Users)
- item_name (string)
- item_price (float)
- item_description (text)
- date_posted (datetime)

Projects Created

1. Iris Classifier:

- Developed a machine learning model to classify iris flowers based on petal and sepal measurements using scikit-learn.

2. Learning Management System (LMS):

- Built a full-stack web application for managing courses, assignments, and student progress with Node.js, Express, and MongoDB.

3. DCGAN for Image Generation:

- Created a deep convolutional generative adversarial network (DCGAN) model in Keras to generate synthetic MNIST images.

GitHub: <https://github.com/Quest4Tech-dev>